LungCT AI | CLINICAL NOISE ASSESSMENT REPORT

Date: 2026-07-26 10:56:00
System: v1.0.0 (U-Net++ model)

Case / Scan ID:	LCT-32672	**Patient Name:**	ANONYMOUS (Clinical Study)
Original File Name:	upscalemedia-transformed.png	**Analysis Protocol:**	U-Net++ Noise Segmentation
Scan Modality:	Computed Tomography (CT)	**Report Status:**	**COMPLETED (AI-Generated)**

CT Scan Imagery Analysis

Figure 1: Original CT Scan
Input diagnostic image

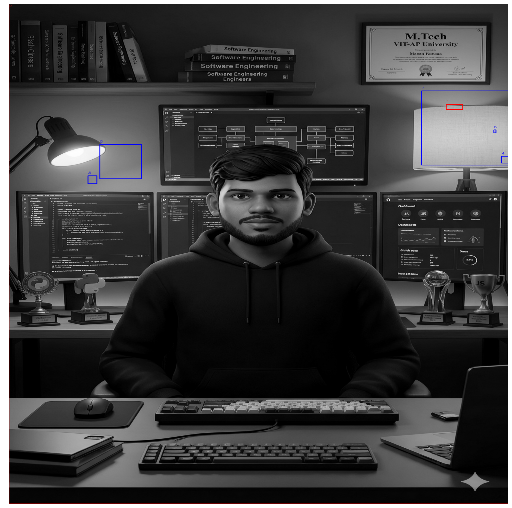

Figure 2: Annotated CT Scan
Red: Gaussian Noise | Blue: Poisson Noise

Quantitative Noise Classification & Severity Metrics

Noise Classification	Pixel Count	Area Coverage (%)	Severity Classification
Gaussian Noise (Electronic)	3,969,172	93.12%	**Critical**
Poisson Noise (Photon Shot)	76,702	1.80%	**Mild**
Clean Regions (Signal)	216,334	5.08%	Normal / Target
TOTAL NOISE (Artifacts)	**4,045,874**	**94.92%**	**Critical**

Clinical Interpretation & Recommendations

Model Findings: CRITICAL noise levels are present in the CT scan, affecting 94.92% of the pixel space. Gaussian noise is 93.12% (Critical) and Poisson noise is 1.80% (Mild). The signal is severely compromised, rendering the scan clinically unreliable for accurate diagnostic interpretation.
Denoising Recommendations: This scan should be flagged as sub-diagnostic. Recommend immediate patient rescan with adjusted dosage/calibration parameters, or review of the imaging hardware for systemic sensor noise.

Radiologist Review Block	AI System Validation Specs
______________________________ Reviewing Radiologist Signature	**Model Architecture:** U-Net++ (Nested UNet) **Dice Score:** 0.9886 | **IoU:** 0.9778 **Precision:** 0.9899 | **Recall:** 0.9875

Disclaimer: This report is automatically generated by an artificial intelligence model (U-Net++) for multi-noise classification and severity assessment. It is developed as a Software Engineering Capstone Project. The findings should be reviewed and verified by a licensed radiologist or healthcare professional before clinical decisions are made. All results are for research and educational validation purposes only.